
Molecule Heterogeneity Limits Probe Compression in Curvature-Supervised Potentials

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Abstract

Reducing random directions within a curvature-supervision update also changes which molecules contribute parameter gradients. We analyze this trade-off with stratified mode coupling: force-residual strata select molecules with exact compensation, while geometry-based sign probes are coupled through randomized Hadamard rows. We prove unbiasedness and derive a gradient-variance decomposition containing an irreducible molecule-selection term and a signed alignment term. An exact counterexample shows that zero scalar-loss variance can coexist with increased gradient variance; more generally, nontrivial cross-molecule coupling cannot improve every curvature-gradient problem. Twenty 80-epoch ANI runs use five matched seeds and complete Test, IRC and NMS evaluation. Nominal twofold and fivefold allocation realize $1.9999\times$ and $4.9998\times$ probe reductions, pass 7 and 6 of nine predeclared accuracy tests, and have maximum pooled gradient-variance ratios of 1.63 and 4.61 relative to Gaussian projected Hessian learning. Neither budget establishes all-cell accuracy non-inferiority. Full-gradient audits, exact molecule-gradient diagnostics and probe-law controls distinguish direction-level variance reduction from the cost of molecule heterogeneity. The measured regime remains below the published converged accuracy and provides no automatic transfer claim to late-convergence or larger-system training.

1 Curvature supervision has two sources of sampling noise

An inexpensive unbiased loss need not give an inexpensive stochastic gradient. For curvature supervision, this distinction arises twice: a random direction approximates a matrix norm within each molecule, and selecting molecules approximates the mean over a minibatch. Reducing the first source of noise does not automatically pay for increasing the second. This matters when energies and forces are already available for a full minibatch but higher derivatives are the expensive part of an update.

Projected Hessian learning (PHL) estimates squared Hessian error with one Hessian–vector product (HVP) per molecule [Rodriguez et al., 2026]. Phonon fine-tuning likewise makes curvature supervision practical by sampling Hessian columns [Koker et al., 2026]. Both motivate a finer allocation question: can an update retain all energy and force information while evaluating curvature on only some of its molecules? Exact compensation removes selection bias, but the resulting gradient noise depends on variation in the curvature gradients of the omitted molecules. A good scalar trace estimate alone cannot answer this question.

We study this issue through *stratified mode coupling* (SMC). A geometry-only spring model supplies a basis for sign probes. Molecules are stratified using their already computed force residuals; one molecule represents each stratum with its exact sampling weight. Randomized Hadamard rows couple directions across the selected molecules. This construction provides a concrete way to separate changes in the probe law, selection, and cross-molecule covariance. Our central theoretical result decomposes its parameter-gradient variance into a selection term, a within-molecule directional term, and a signed cross-molecule term. The selection term is an irreducible lower bound for this family,

and the coupling term is beneficial only when gradient matrices are aligned. This explains why large savings are not a consequence of orthogonality or unbiasedness alone.

Relation to variance reduction. Hutchinson estimation and its analysis establish how the fourth moments of a probe control trace variance [Hutchinson, 1990, Avron and Toledo, 2011]. Hutch++ constructs a low-rank control variate, and XTrace uses exchangeability to reuse matrix–vector products more efficiently [Meyer et al., 2021, Epperly et al., 2024]. Their primary object is a single implicit matrix; in curvature learning, every configuration has its own matrix and every parameter differentiates that matrix differently. Orthogonal Monte Carlo and structured random features provide powerful mechanisms for coupling samples [Yu et al., 2016, Lin et al., 2020]; hierarchical probing exploits matrix locality [Stathopoulos et al., 2013]. We do not claim orthogonal probes, sign probes, stratification, or inverse-probability compensation as new primitives.

Importance sampling for stochastic optimization allocates examples according to their gradient magnitudes or tractable proxies [Needell et al., 2014, Katharopoulos and Fleuret, 2018]. That work highlights why loss magnitude is an imperfect proxy for gradient utility. Here, the force residual used for stratification is especially inexpensive, but it is not an oracle for the Hessian-loss gradient. Our contribution is the heterogeneity-sensitive gradient analysis of coupled curvature probes, together with a controlled study that measures this mismatch directly. The comparison includes the published PHL reference, an executable-code reproduction, a full-allocation probe-law control, and two explicitly compensated allocation rates.

2 Stratified mode coupling

Let a minibatch contain B configurations. Configuration i has N_i atoms, $d_i = 3N_i$ Cartesian coordinates, and Hessian residual $D_i = \tilde{H}_i - H_i$. Write

$$A_i = D_i^\top D_i / d_i^2, \quad \ell_i = \text{tr } A_i, \quad \mathcal{L}_H = B^{-1} \sum_{i=1}^B \ell_i. \quad (1)$$

The training objective retains the reference implementation’s normalization:

$$\mathcal{L} = B^{-1} \sum_i \left[\frac{(\tilde{E}_i - E_i)^2}{\sqrt{N_i}} + 0.30 \frac{\|\tilde{F}_i - F_i\|^2}{d_i} \right] + 0.09 \hat{\mathcal{L}}_H. \quad (2)$$

Every molecule contributes to the energy and force terms at every update. Only the estimator of $\mathcal{L}_H$ changes.

2.1 Directions and allocation

For each input geometry, construct a central-spring matrix

$$S_i = \sum_{a < b: r_{ab} \leq 1.35(\rho_a + \rho_b)} r_{ab}^{-2} u_{ab} u_{ab}^\top. \quad (3)$$

Here u_{ab} contains $+(R_a - R_b)/r_{ab}$ at atom a , its negative at atom b , and zero elsewhere. The radii in angstroms are the fixed design values (0.31, 0.76, 0.71, 0.66) for (H, C, N, O). Numerically coincident pairs with $r_{ab} < 10^{-8}$ Å are omitted. Let Q_i be the orthonormal eigenvectors returned by symmetric eigendecomposition, ordered by ascending eigenvalue. All modes, including the null space, are retained. No physical Hessian, energy, force label, or learned model is used to construct Q_i . The basis is cached from the input geometry; its construction time and storage are charged separately.

At each update, sort the molecules by $s_i = \|\tilde{F}_i - F_i\|/\sqrt{d_i}$, with stable tie breaking. Detach this score from autodifferentiation. Partition the sorted sequence into successive strata G_g of size r , except the last stratum, which may be smaller. Draw I_g uniformly and independently from each stratum. Set $K_g = |G_g|$, $a_g = K_g/B$, and $P = \lceil B/r \rceil$. We study $r = 2$ (SMC-2) and $r = 5$ (SMC-5); there is no warm-up, annealing, or unlogged allocation probe.

To couple directions, pad the selected configurations to dimension $D = \max_g d_{I_g}$ and set $L = 2^{\lceil \log_2 D \rceil}$. Randomly permute the P selected molecules into blocks of at most L . Within each block,

draw distinct rows of an $L \times L$ Hadamard matrix, uniformly without replacement, and multiply every row by the same fresh independent Rademacher sign vector. Different blocks use independent signs and row permutations. Restrict row w_g to its first d_{I_g} coordinates and set $v_g = Q_{I_g} w_g$. The loss estimate is

$$\hat{\mathcal{L}}_H = \sum_{g=1}^P a_g \frac{\|D_{I_g} v_g\|^2}{d_{I_g}^2}. \quad (4)$$

Each selected molecule requires one model HVP and its reference product. A separate coordinate graph for the selected subbatch ensures that omitted molecules do not undergo a hidden HVP. The full-allocation control, Mode-full, uses independent Q_i -rotated sign probes on every molecule. Gaussian PHL uses independent standard normal probes, as in the authors’ notebook.

2.2 Exact unbiasedness

Proposition 1. *Conditional on the current parameters, minibatch, and input bases, Eq. (4) is an unbiased estimator of $\mathcal{L}_H$. Its autodifferentiated parameter gradient is unbiased wherever the model’s required derivatives exist.*

Proof. Every marginal Hadamard row multiplied by independent signs has $\mathbb{E}[w_g w_g^\top] = I$. Truncation preserves this identity on the active coordinates, and orthogonality of Q_i gives $\mathbb{E}[v_g v_g^\top \mid I_g = i] = I_{d_i}$. Consequently $\mathbb{E}[\|D_i v_g\|^2 / d_i^2 \mid I_g = i] = \ell_i$. The dependence between different rows does not change these marginal expectations. Since $\Pr(I_g = i) = 1/K_g$,

$$\mathbb{E} \hat{\mathcal{L}}_H = \sum_g \frac{K_g}{B} \frac{1}{K_g} \sum_{i \in G_g} \ell_i = \mathcal{L}_H. \quad (5)$$

The same argument applies to each parameter derivative of the quadratic form. The bases depend only on coordinates. Although force-based strata depend on the current parameters, at every fixed parameter value the compensated conditional mean of the sampled derivatives is the derivative of the full minibatch loss. No derivative of a selection probability is needed. This statement holds away from derivative discontinuities of the piecewise-smooth reference network and neighbor cutoffs, as does its ordinary force/Hessian training implementation. $\square$

2.3 Gradient variance and its heterogeneity floor

The relevant quadratic form for parameter θ_k is not A_i but

$$C_{ik} = Q_i^\top (\partial_{\theta_k} A_i) Q_i = d_i^{-2} Q_i^\top [(\partial_k D_i)^\top D_i + D_i^\top (\partial_k D_i)] Q_i. \quad (6)$$

Pad these symmetric matrices with zeros to the common probe dimension. Let $g_i = (\text{tr } C_{ik})_k$, let C_i° collect their off-diagonal entries, and use the Frobenius inner product over both matrix entries and parameters. Define

$$\sigma_i^2 = 2\|C_i^\circ\|_F^2, \quad \tau_{ij} = 2\langle C_i^\circ, C_j^\circ \rangle_F. \quad (7)$$

The trace covariance of one sign-probed gradient is σ_i^2 . For independent Gaussian PHL it is instead $2\sum_k \|C_{ik}\|_F^2$. Thus reducing scalar loss variance does not establish a corresponding reduction of parameter-gradient variance.

An exact counterexample. Consider two molecules with $A_1(\theta) = I_2 + \theta C$ and $A_2(\theta) = I_2 - \theta C$, where $C = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Both loss matrices are positive definite for $|\theta| < 1$ and admit Hessian-residual square-root factorizations. At $\theta = 0$, every sign probe gives the exact scalar loss, so its variance is zero. With the two orthogonal rows (z_1, z_2) and $(z_1, -z_2)$, however, the averaged parameter gradient is $2z_1 z_2$, with variance four. Independent sign probes give gradient variance two. Thus exact scalar-loss estimation can coexist with a doubling of gradient noise. The issue is the opposite alignment of the two derivative matrices, not loss bias.

Conditional on the selected molecules, let block sizes be n_b and define

$$\kappa = \frac{\sum_b n_b(n_b - 1)}{P(P - 1)(L - 1)} \quad (P > 1), \quad \kappa = 0 \quad (P = 1). \quad (8)$$

Then the exact conditional directional variance is

$$V_{\text{dir}}(I) = \sum_g a_g^2 \sigma_{I_g}^2 - \kappa \sum_{g \neq h} a_g a_h \tau_{I_g I_h}. \quad (9)$$

To see this, expand two quadratic forms. Averaging the shared signs removes every term except identical unordered coordinate pairs. For distinct Hadamard rows, the surviving row-product expectation is $-1/(L-1)$. Random block assignment multiplies this by the probability that two molecules share a block. Diagonal terms are deterministic. This proves Eq. (9) without assuming that gradient matrices are positive semidefinite.

The size of the possible benefit is explicit. Set $Z_g = a_g C_{I_g}^\circ$ and, when the denominator is positive, define the alignment statistic

$$\rho = \frac{\|\sum_g Z_g\|_F^2 - \sum_g \|Z_g\|_F^2}{(P-1) \sum_g \|Z_g\|_F^2}, \quad -\frac{1}{P-1} \leq \rho \leq 1. \quad (10)$$

Equation (9) becomes $V_{\text{dir}} = 2 \sum_g \|Z_g\|_F^2 [1 - \kappa(P-1)\rho]$. Thus coupling removes a fraction $\kappa(P-1)\rho$ of independent directional variance. If $P \leq L$, even perfectly aligned equal-weight matrices leave a fraction $(L-P)/(L-1)$: incomplete orthogonal blocks cannot cancel all directional noise. In the idealized equal-stratum case with zero selection variance, if a full-allocation probe law has variance ratio q relative to Gaussian PHL, reduction by r requires a coupling gain of at least $1 - 2/(rq)$ to stay within twice PHL variance. Positive selection variance makes this requirement stricter.

Proposition 2 (No uniform benefit from cross-molecule coupling). *For fixed marginal probe laws, any coupling with a nonzero covariance between two quadratic forms increases parameter-gradient variance for some locally positive-definite loss matrices, relative to independent probes with the same marginals.*

Proof. Choose symmetric C_i, C_j whose coupled quadratic forms have nonzero covariance, and set every other derivative matrix to zero. The difference from independent-gradient variance is twice that covariance times the positive averaging weights. Replacing C_j by $-C_j$ reverses its sign, so one choice makes the difference positive. Both choices are realized by $A_i(\theta) = I + \theta C_i$ and $A_j(\theta) = I + \theta C_j$ for sufficiently small $|\theta|$. These matrices are positive definite and can be factored as $D^\top D/d^2$ with symmetric $D = d\sqrt{A}$, itself the Hessian of a quadratic energy. The obstruction therefore also occurs for valid local curvature losses, rather than only arbitrary signed scalar estimands. $\square$

The law of total variance now gives

$$\text{tr Cov}(\nabla_\theta \hat{\mathcal{L}}_H) = V_{\text{sel}} + \mathbb{E}_I V_{\text{dir}}(I), \quad (11)$$

$$V_{\text{sel}} = \frac{1}{B^2} \sum_g K_g \sum_{i \in G_g} \|g_i - \bar{g}_g\|^2, \quad \bar{g}_g = K_g^{-1} \sum_{i \in G_g} g_i. \quad (12)$$

The expectation in Eq. (11) also accounts for any selection-dependent change in the padded dimension. Because conditional variance is nonnegative, V_{sel} is an irreducible floor. Orthogonal coupling is most effective when the off-diagonal gradient matrices align; negative alignment can increase variance. If all gradient matrices are identical and a block uses every Hadamard row, its directional variance vanishes exactly. Neither premise holds automatically for chemically diverse molecules. Force sorting attempts to reduce Eq. (12), but low force residuals do not imply similar Hessian gradients, particularly near stationary structures.

For equal strata with $B = rP$, the independent-direction control gives a particularly useful identity. If V_{mode} is the variance of one independent mode-sign probe on every molecule, then

$$V_{\text{ind}}(r) = rV_{\text{mode}} + V_{\text{sel}}(r). \quad (13)$$

Indeed, averaging $a_g^2 \sigma_{I_g}^2$ over the uniform selections gives $(r/B^2) \sum_i \sigma_i^2$. This permits a full-minibatch estimate of the selection component from independent full- and reduced-allocation audits, without forming every molecule’s exact Hessian gradient. The signed difference $V_{\text{SMC}} - V_{\text{ind}}$ separately measures the coupling contribution. Finite-sample estimates of these differences can fluctuate around their population values.

This analysis predicts a limited compression regime: replacing Gaussian probes by sign probes can remove diagonal noise, whereas increasing r multiplies the independent directional contribution and adds molecule selection noise. A large reduction requires measurable cross-molecule alignment or an accurate proxy for curvature gradients, not just an isotropic direction distribution.

3 Controlled reactive-chemistry experiments

Data and implementation. We use the supplied OpenREACT CHON energy, force and Hessian data [Rodriguez et al., 2025]. The local files contain 35,087 RTP, 34,248 IRC and 62,527 NMS configurations. Each supplied file pair has the same SHA-256 hash; one copy of each dataset is used. A seeded Python shuffle gives the RTP 80/10/10 split: 28,069 training, 3,508 validation, and 3,510 test configurations. IRC and NMS are used exclusively for evaluation and the post-training diagnostic audits. No geometry is reassigned between these roles. The five matched seeds are 7289038–7289042; splits vary by seed but are identical between estimators for each seed. The original split indices are not distributed, so these reproduce the published split *procedure*, not a verified identical held-out membership.

The executable reference distributed with PHL is an ANI potential [Smith et al., 2017, Gao et al., 2020]; the repository mentions a separate HIP-NN integration but does not supply that training pipeline. We execute the ANI model-definition cells directly: four 384–256–64–256–1 atomic networks, CELU(0.1), radial and angular cutoffs of 5.2 and 3.5 Å, and double precision. The reference utility module provides Hessian-aware padding and exact evaluation Hessians. The HVP, loss normalizations and separate optimizers for weights and biases follow the notebook. The work is an ANI study, not a HIP-NN reproduction. Batched Gaussian generation preserves the reference probe law while changing random-number consumption; the single-device DataParallel wrapper is removed. Published timing differences therefore combine hardware and implementation effects and do not measure an allocation gain.

Optimization and scope. All estimators use the entire RTP training split, nominal minibatches of 400, fixed cached minibatch order, and 80 epochs. The final minibatch has 69 configurations; all probe counts use its actual size. AdamW updates weights and SGD updates biases with the notebook’s learning rates of 10^{-3} ; both have ReduceLROnPlateau schedules with factor 0.5, patience 100, and threshold zero. These executable learning rates differ from the 10^{-4} stated in the paper’s supplement. Self energies are least-squares fitted on the training split only, rather than fitted before splitting as in the notebook. The loss weights are exactly those in Eq. (2). The checkpoint with the lowest validation energy RMSE is evaluated. There is no test-based model selection or early termination of the announced epoch budget.

To afford the paired estimators and gradient audits, the 80-epoch schedule is shorter than the original paper’s late-convergence regime, which discusses stabilization after approximately 700 epochs. The published PHL row remains visible as an external accuracy anchor; parity with our implementation must not be interpreted as parity with that converged reference. Numerical gaps are reported explicitly in the results. Computation uses one visible NVIDIA H200 GPU, the hardware actually exposed by the environment, with four CPU threads, rather than the A6000 used for the published timings. Within-study settings are identical for all estimators. Auxiliary CPU-only pilot checks overlap a subset of GPU training epochs; the epoch timings describe those executed runs. A separate phase profile is performed after timed training and the CPU checks finish, to assess overhead without that contention.

Selection, statistics, and measurement. A preliminary Gaussian model was trained for 400 updates. Its two RTP validation minibatches compare probe laws, coupling, and allocation factors with 32 gradient draws each and an independent 64-probe oracle. Only these validation measurements informed the primary choice $r = 2$; $r = 5$ remains a prespecified stress test. The final protocol was written to disk before any final-seed test evaluation. The same-probe-count control, Mode-full, uses independent mode-basis signs on every molecule.

Accuracy is assessed by a prespecified one-sided paired non-inferiority test. For each of the nine dataset/property cells, let $z_s = \log(\text{RMSE}_{s,\text{method}}/\text{RMSE}_{s,\text{PHL}})$. The cell passes if

$$\bar{z} + t_{0.95,4} s_z / \sqrt{5} < \log(1.05). \quad (14)$$

Thus the allowed degradation is 5% in RMSE, the paired experimental unit is the training seed, and the confidence bound is one-sided 95%. All nine cells must pass; an average over properties cannot establish non-inferiority. This intersection–union rule needs no multiplicity adjustment for the all-cell claim. It does not assert equivalence. We report arithmetic mean and sample standard deviation of RMSE, all individual seed values, the paired confidence bounds, and the one-sided p -values.

Gradient audits freeze both the PHL and SMC-2 trajectories at epochs 1, 20, and 80. At each checkpoint, two fixed minibatches of 400 molecules per dataset and seed are evaluated with 24 fresh draws of each estimator. The statistic is the trace of the sample covariance of the *entire* unweighted Hessian-loss parameter gradient; no projection or per-parameter renormalization is used. At a fixed minibatch, the energy/force gradient is deterministic and the Hessian coefficient scales covariance by 0.09^2 for every estimator. The conditional variance ratio therefore also equals that of the full objective’s gradient. An independent 64-probe Gaussian mean supplies a many-probe oracle with its Monte Carlo standard error. Gradients are computed in float64, archived in float32, and covariance traces accumulated in float64. Additional exact full-Hessian gradients check the oracle at the epoch-80 checkpoint of the first seed. Diagnostic gradients never choose an update and are accounted separately from training probes.

We also estimate total minibatch-plus-probe Hessian-gradient variance by adding the between-minibatch covariance of the oracle means, correcting for their finite Monte Carlo noise. Conditional variance remains the primary diagnostic so that the noise introduced by allocation is directly visible.

The dataset-level variance ratio divides pooled conditional variances, averaging the same seeds, checkpoints, minibatches, and both model paths in numerator and denominator. Confidence intervals resample the five seed clusters. A point ratio above 2 fails the announced threshold even if its interval crosses 2; stage-specific excesses are also reported. Final RMSEs use all Test, IRC and NMS configurations and exact Hessians. Validation curves use the full validation split for energy and force and its fixed first 128 shuffled members for exact Hessian RMSE.

Accounting. The primary resource statistic is the number of molecule–direction HVP evaluations used by an update, averaged over *all* training steps. One batched autograd invocation on P molecules therefore counts as P probes, not one. Basis construction uses no HVP or curvature labels. Every minibatch index, selection decision and probe count is saved. The complete float64 probe vectors are archived by replaying each run’s isolated random generator, checking every sampled molecule and its saved terminal generator state; replay computes no model HVP. GPU-synchronized epoch timings include allocation, data movement and optimization, and exclude validation, audit computation and disk logging for all methods. Preprocessing and diagnostic costs are reported separately.

4 Accuracy and gradient noise across training

The comparison comprises twenty trained models and complete evaluation on all three datasets. Table 1 places the published PHL reference beside the matched implementations. Our PHL RMSEs are 1.33–1.89 times the published values across the nine cells. This gap is consistent with the explicitly shorter optimization regime; the learning-rate and split-membership differences in Section 3 also prevent attributing the entire gap to duration. The published row is an external anchor, not the control used for inference.

Figure 1 shows the optimization trajectories rather than only the selected checkpoints. Figure 2 retains the variation between seeds, and Table 5 supplies every numerical outcome. Energy-based checkpoint selection is the same for all methods; force and Hessian results are evaluated at that checkpoint, rather than selecting a separate optimum for each property.

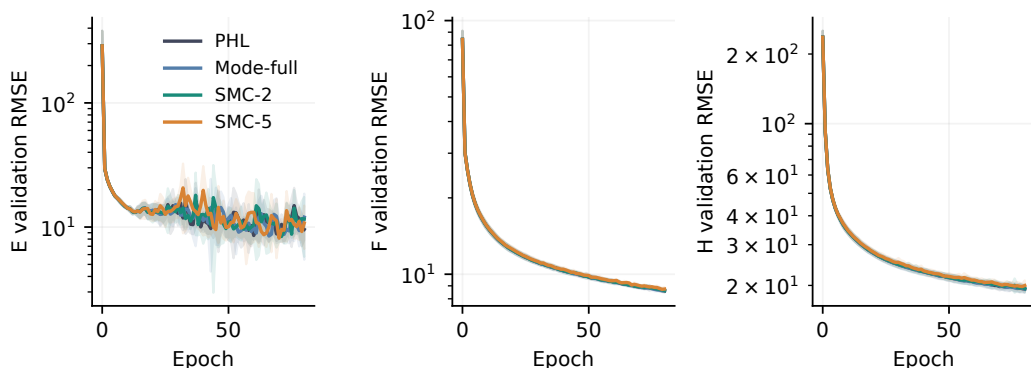


Figure 1: Validation convergence across all five seeds: arithmetic means and sample-standard-deviation bands. Energy and force use the full validation split; Hessian RMSE uses the fixed 128-molecule validation subset. The vertical axes are logarithmic. Units are kcal mol^{-1} , $\text{kcal mol}^{-1}\text{\AA}^{-1}$, and $\text{kcal mol}^{-1}\text{\AA}^{-2}$, respectively. No final-test value selects an epoch or estimator.

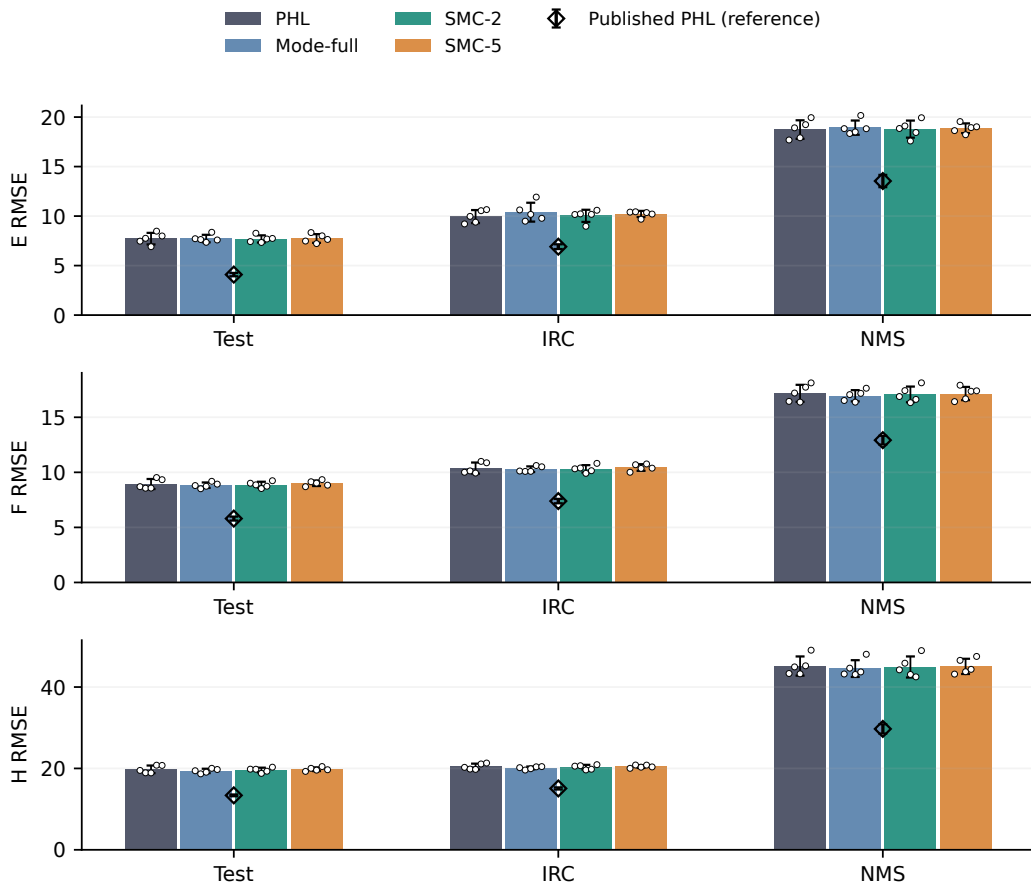


Figure 2: Final energy, force and Hessian RMSEs, in the dataset grouping of [Rodriguez et al. \[2026\]](#), Figure 3(a). Bars and whiskers are our five-seed means and sample standard deviations; individual seed outcomes are overlaid. Open diamonds show published PHL means, with approximate standard deviations digitized from the source figure. Units match Figure 1. Published and reimplemented values describe different optimization durations and are not paired observations.

Table 1: Complete accuracy and resource comparison. Reimplemented values are mean $\pm$ sample standard deviation across the five paired seeds. E, F and H units are kcal mol⁻¹, kcal mol⁻¹Å⁻¹ and kcal mol⁻¹Å⁻². Every published RMSE is the purple PHL bar label in [Rodriguez et al. \[2026\]](#), Figure 3(a), p. 12; published standard deviations are approximate, digitized from that figure’s vector error bars (marked d). The 13.34 s runtime is Figure 4, p. 16; $B = 400$ and one probe per molecule are Supplement B, p. 26. The published probe entry is nominal; the source does not tabulate a short-batch-adjusted average. Their A6000 timings are external references. Our timings use one H200. The parity column gives the predeclared one-sided 5% non-inferiority test on Test E only; Table 2 gives every cell. The last column is the maximum pooled dataset-specific conditional gradient-variance ratio, with all datasets and stages in Table 3.

Method	Probes / minibatch step (avg)	Per-epoch wall-clock (s)										Accuracy-parity test	
			Test E RMSE	Test F RMSE	Test H RMSE	IRC E RMSE	IRC F RMSE	IRC H RMSE	NMS E RMSE	NMS F RMSE	NMS H RMSE	vs. reimpl. PHL (Test E)	Gradient-var ratio (your / reimpl. PHL)
∞	PHL (paper as reported) ^d	B (nominal 400)	4.09	5.80	13.38	6.91	7.39	15.08	13.54	12.92	29.71	—	—
	PHL (reimplemented, 5 seeds)	395.34	2.73 ±0.06	7.72 ±0.58	8.94 ±0.46	19.77 ±0.94	9.95 ±0.66	10.39 ±0.50	20.46 ±0.72	18.74 ±0.94	17.18 ±0.77	45.16 ±2.36	1.00
	Mode-full	395.34	2.82 ±0.09	7.73 ±0.38	8.83 ±0.25	19.39 ±0.54	10.39 ±0.95	10.28 ±0.26	20.11 ±0.35	18.93 ±0.72	16.95 ±0.51	44.54 ±2.06	$p = 0.041$ pass
	SMC-2	197.68	4.22 ±0.14	7.69 ±0.36	8.88 ±0.27	19.60 ±0.58	10.02 ±0.61	10.32 ±0.33	20.32 ±0.57	18.78 ±0.86	17.08 ±0.71	44.93 ±2.59	$p = 0.082$ fail
	SMC-5	79.07	3.68 ±0.14	7.74 ±0.44	9.01 ±0.25	19.80 ±0.46	10.21 ±0.31	10.44 ±0.30	20.47 ±0.35	18.87 ±0.49	17.15 ±0.60	45.08 ±1.87	$p = 0.070$ fail

Table 2: All-cell accuracy assessment. Each entry is the one-sided 95% upper bound on the paired relative RMSE change (percent); the bound must be below 5%. A superscript $\dagger$ marks failure. The complete p -values and five paired log ratios are in `logs/summary.json`.

Method	Dataset	Energy	Force	Hessian
Mode-full	Test	4.61	1.57	0.38
Mode-full	IRC	15.11 $\dagger$	1.36	0.32
Mode-full	NMS	6.24 $\dagger$	0.30	−0.08
SMC-2	Test	6.38 $\dagger$	4.16	3.48
SMC-2	IRC	6.45 $\dagger$	3.66	3.07
SMC-2	NMS	4.09	2.81	2.70
SMC-5	Test	5.76 $\dagger$	5.77 $\dagger$	4.47
SMC-5	IRC	9.22 $\dagger$	4.66	3.71
SMC-5	NMS	4.66	2.99	2.39

Accuracy decisions. At the predeclared 5% non-inferiority margin, SMC-2 passes 7 of nine cells and SMC-5 passes 6 of nine. Mode-full passes 7 cells. Table 2 reports every dataset/property confidence bound. A failed bound means that non-inferiority was not established; it does not by itself prove a degradation exceeding 5%. No difference test is substituted after observing these outcomes.

Direct gradient measurements. The 180 frozen-minibatch audits measure all 527,620 parameter-gradient components. Every estimator in a comparison sees exactly the same model and molecule identities. Table 3 gives the conditional variance ratios, including the independent-direction SMC-2 ablation. Mode-2 independent is evaluated only on frozen checkpoints; it does not have a separate accuracy-training run. The maximum pooled ratio across datasets is 1.63 for SMC-2 and 4.61 for SMC-5. Figure 3 separates the three training stages. SMC-2 has pooled ratios 1.62, 1.63 and 1.61 on Test, IRC and NMS; each is below two. Its ratios also remain below two at every audited epoch on both model trajectories after pooling the matched seeds and minibatches. SMC-5 has ratios 4.61, 4.52 and 4.52, exceeding the threshold on every dataset. The pooled decision is not a uniform minibatch bound: 10 of the 180 individual 24-draw SMC-2 ratios exceed two (five Test and five IRC; range 1.17–2.35). These descriptive block estimates have Monte Carlo noise. Every individual SMC-5 block ratio exceeds two.

The 64-probe oracle and the supplemental exact full-Hessian gradients are diagnostic computations, never update inputs. Across the 180 audits, the oracle’s estimated standard-error norm is 1.99–9.33% of its mean gradient norm. In the six exact checks, relative gradient errors are 3.20–7.11%, or 1.02–1.48 times their estimated Monte Carlo error norms. Thus the finite oracle is not treated as an exact gradient; the unbiasedness claim rests on the proof and exact finite-probe check.

Adding the estimated between-minibatch term gives Table 4. Because this term is shared by unbiased estimators, it moves ratios toward one. It does not remove the extra conditional noise introduced by allocation. The two-minibatch estimate of that term is less precise than the repeated-direction comparison; it is included to distinguish the two meanings of gradient variance.

Actual probe budgets. PHL uses 395.34 probes per update averaged over the full run. SMC-2 uses 197.68 and SMC-5 uses 79.07, giving reductions of $1.9999\times$ and $4.9998\times$, respectively. The names refer to nominal stratum sizes; the smaller final minibatch and ceiling in $P = \lceil B/r \rceil$ make the realized ratios slightly below two and five. There is no initial dense-probe phase omitted from these averages. Figure 6 shows every update’s allocation count.

Table 3: Conditional parameter-gradient variance relative to Gaussian PHL, on the same models and minibatches. Pooled entries give ratios with 95% seed-cluster bootstrap intervals; stage columns pool both paths and their minibatches at the named epoch. A superscript $\dagger$ marks an unrounded point estimate above 2, exceeding the declared threshold even if its interval crosses 2.

Estimator	Dataset	Pooled [95% CI]	Epoch 1	Epoch 20	Epoch 80
Mode-full	Test	0.67 [0.65, 0.69]	0.66	0.79	0.77
Mode-full	IRC	0.66 [0.64, 0.67]	0.65	0.76	0.72
Mode-full	NMS	0.64 [0.62, 0.66]	0.67	0.63	0.55
SMC-2	Test	1.62 [1.57, 1.67]	1.60	1.82	1.84
SMC-2	IRC	1.63 [1.58, 1.66]	1.61	1.81	1.75
SMC-2	NMS	1.61 [1.57, 1.64]	1.64	1.58	1.47
SMC-5	Test	4.61 † [4.45, 4.78]	4.57 †	5.11 †	5.17 †
SMC-5	IRC	4.52 † [4.35, 4.67]	4.48 †	4.99 †	4.75 †
SMC-5	NMS	4.52 † [4.39, 4.70]	4.58 †	4.48 †	4.31 †
Mode-2 independent	Test	1.62 [1.57, 1.66]	1.60	1.85	1.86
Mode-2 independent	IRC	1.62 [1.58, 1.66]	1.60	1.81	1.81
Mode-2 independent	NMS	1.61 [1.58, 1.64]	1.65	1.58	1.48

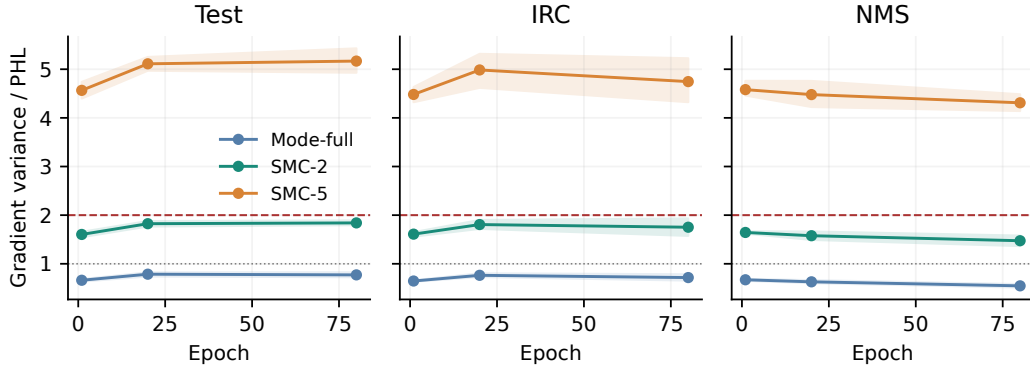


Figure 3: Conditional parameter-gradient variance relative to Gaussian PHL at epochs 1, 20, and 80. Ratios pool the same minibatches and the PHL and SMC-2 model paths at each epoch. Bands are 95% bootstrap intervals resampling the five paired seed clusters. The dashed line marks the declared ratio of two; the dotted line denotes PHL variance. The connecting segments guide the eye and do not represent audits at unmeasured epochs.

Table 4: Total minibatch-plus-probe Hessian-gradient variance ratios, including the between-minibatch term estimated from the independent oracle means. Finite-oracle mean noise is subtracted before pooling, with negative corrected between-minibatch estimates clipped to zero. Intervals resample paired seed clusters. The conditional ratios in Table 3 remain the primary estimator diagnostic.

Estimator	Test	IRC	NMS
Mode-full	0.75 [0.73, 0.78]	0.74 [0.73, 0.75]	0.74 [0.72, 0.76]
SMC-2	1.47 [1.43, 1.51]	1.47 [1.44, 1.51]	1.44 [1.40, 1.49]
SMC-5	3.70 [3.56, 3.85]	3.65 [3.50, 3.82]	3.58 [3.33, 3.88]

Table 5: Individual model outcomes. All five paired training seeds are shown; the selected checkpoint minimizes validation energy. These seed-level RMSEs, not the individual geometries, are the units of the paired accuracy tests. Units match Table 1.

Method	Seed	Epoch	Test E	Test F	Test H	IRC E	IRC F	IRC H	NMS E	NMS F	NMS H
PHL	7289038	72	7.47	8.70	19.51	9.21	10.01	20.26	17.69	16.44	43.33
PHL	7289039	75	7.76	8.56	18.92	9.97	10.13	19.84	18.91	17.20	44.95
PHL	7289040	77	6.91	8.57	18.91	9.39	9.93	19.76	17.91	16.38	43.25
PHL	7289041	58	8.48	9.53	20.79	10.54	10.99	21.10	19.23	17.73	45.21
PHL	7289042	62	7.99	9.33	20.74	10.65	10.87	21.32	19.94	18.12	49.06
Mode-full	7289038	71	7.71	8.80	19.43	10.62	10.12	20.19	18.83	16.52	43.21
Mode-full	7289039	78	7.63	8.51	18.67	9.49	10.08	19.60	18.34	17.06	44.64
Mode-full	7289040	75	7.35	8.74	19.07	10.17	10.09	19.93	18.50	16.37	43.08
Mode-full	7289041	67	8.36	9.18	20.03	11.92	10.62	20.39	20.16	17.17	43.71
Mode-full	7289042	73	7.59	8.93	19.74	9.78	10.50	20.45	18.82	17.64	48.05
SMC-2	7289038	67	7.43	9.01	19.82	10.16	10.32	20.56	18.85	16.88	44.23
SMC-2	7289039	65	8.27	8.86	19.80	10.22	10.40	20.67	19.10	17.42	45.86
SMC-2	7289040	80	7.33	8.53	18.78	8.97	9.91	19.63	17.59	16.33	43.09
SMC-2	7289041	80	7.68	8.74	19.32	10.18	10.15	19.81	18.44	16.62	42.50
SMC-2	7289042	67	7.74	9.24	20.31	10.59	10.81	20.92	19.93	18.13	48.96
SMC-5	7289038	78	7.48	8.70	19.27	10.40	10.01	20.03	18.63	16.42	43.18
SMC-5	7289039	62	8.34	9.14	20.02	10.43	10.69	20.84	19.54	17.91	46.56
SMC-5	7289040	72	7.22	9.02	19.58	9.68	10.36	20.31	18.22	16.67	43.77
SMC-5	7289041	65	8.01	9.34	20.46	10.35	10.76	20.82	18.94	17.36	44.35
SMC-5	7289042	80	7.64	8.84	19.68	10.21	10.38	20.37	19.02	17.40	47.52

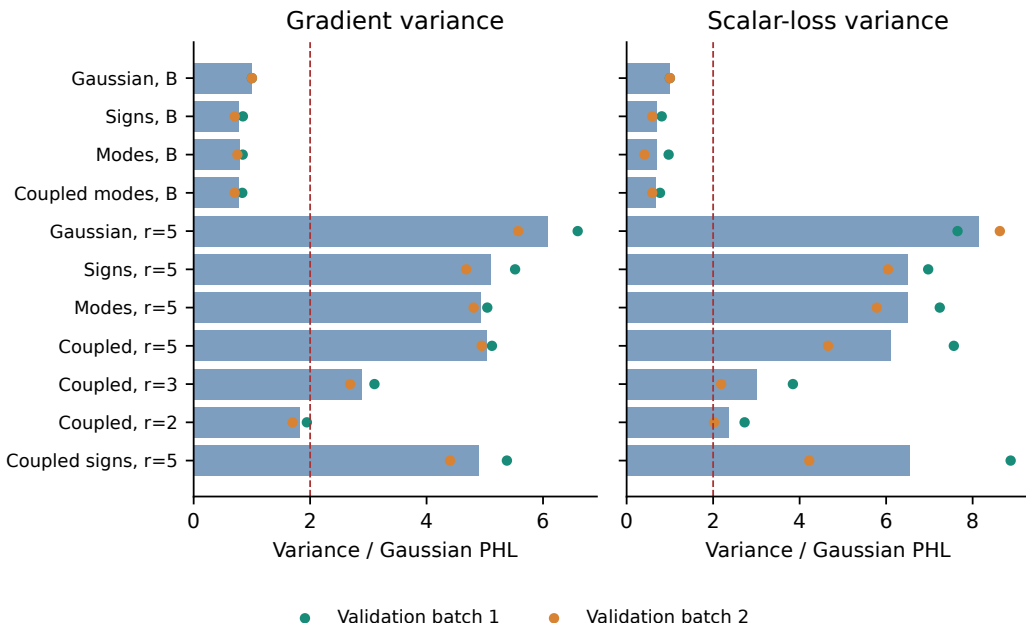


Figure 4: Validation-only component study at the 400-update pilot checkpoint. Each dot is a separate 400-molecule minibatch with 32 repeated gradient measurements; bars average the two ratios. Full allocation separates the probe-law effect from reduced allocation. Reduced sign/mode variants use force strata; reduced Gaussian uses uniform selection. Coupling refers to randomized Hadamard rows. Scalar-loss variance (right) is measured separately from the full parameter-gradient variance (left). These are exploratory measurements, not five-seed final accuracy comparisons.

5 What limits the allocation gain?

5.1 Mechanism analysis: directional noise and molecule heterogeneity

The full-minibatch decomposition in Eq. (13) is directly identifiable from the independent-direction audit. In units of Gaussian PHL variance, twofold allocation has directional contributions 1.34, 1.31 and 1.29 on Test, IRC and NMS. The estimated selection terms are 0.28 [0.25, 0.31], 0.31 [0.26, 0.36] and 0.32 [0.30, 0.35], with 95% seed-cluster bootstrap intervals. Coupling changes the totals by only +0.002, +0.004 and −0.006, and every corresponding interval includes zero. The measurable variance reduction at full allocation therefore comes from the probe law; these data do not detect a consistent additional reduction from coupling.

The pilot isolates the inexpensive change of probe law from actual compression. Mode-full has gradient-variance ratios 0.842 and 0.751 on the two validation minibatches. With fivefold selection, independent mode probes give 5.043 and 4.809; coupling gives 5.123 and 4.946. Thus the point estimates provide no compensating coupling gain at this budget. Even in the favorable zero-selection-noise calculation in Section 2, roughly half the remaining directional variance would have to disappear to make these probe-law gains pay for fivefold compression. The observed values do not approach that requirement.

A separate importance-sampling check varies the force-score exponent and atom-count normalization while retaining a positive uniform probability floor. Even the separately best settings on the two validation minibatches give gradient-variance ratios 4.48 and 4.72 at 80 probes. Thus the tested simple reweightings do not recover the missing variance reduction. Appendix A reports every setting; these diagnostic choices do not revise the frozen training comparison.

Scalar-loss variance can rank the same changes differently. On the second validation minibatch, full-allocation independent and coupled mode probes have scalar-variance ratios 0.411 and 0.592, but gradient ratios 0.751 and 0.704. The exact counterexample establishes that such disagreement

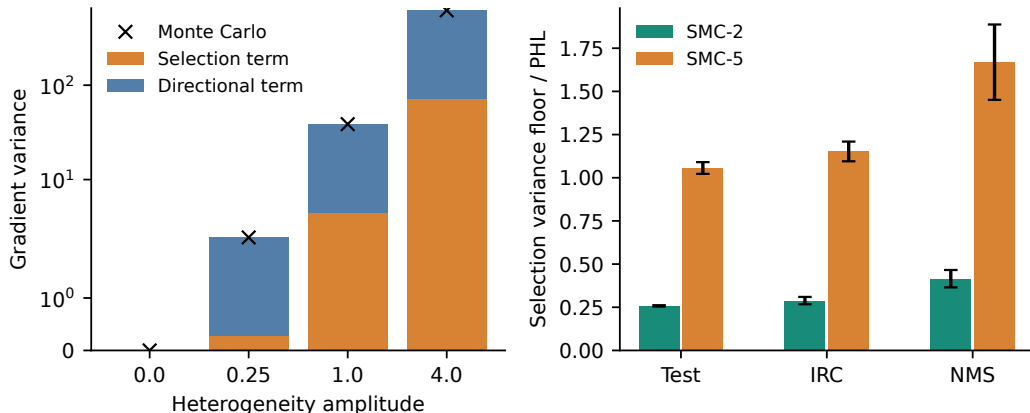


Figure 5: Heterogeneity diagnostics. Left: exact theoretical components and independent Monte Carlo variances for synthetic derivative matrices, using $B = 40$, $r = 5$, dimension eight and five parameters. The heterogeneous perturbation amplitude varies while the coupling construction is fixed. Right: exact molecule-gradient selection floors on 40-molecule diagnostics at epoch 80, normalized by measured Gaussian variance on those same batches; bars show five-seed means and standard deviations. The right panel is an auxiliary small-batch measurement, not the primary 400-molecule variance assessment.

is structural, rather than a claim resting on the precision of this particular pilot comparison. The differentiated residual matrices C_{ik} , rather than the loss matrices A_i , determine optimization noise.

Figure 5 checks the variance decomposition independently of the neural model. Eight synthetic settings use 10,000 draws each; all nonzero predicted variances agree with simulation to within 0.8%. For identical matrices and a complete Hadamard block, cancellation is exact up to floating-point roundoff. Adding molecule-specific derivative variation restores both selection and directional noise. This confirms the mechanism without presuming that chemical configurations exhibit the ideal alignment. An implementation check on two four-atom RTP configurations also enumerates the complete rotated Hadamard direction set and both molecule choices. Its averaged gradient agrees with an independently differentiated full-Hessian loss to relative error 9.74×10^{-16} , verifying the compensation and projected-loss code on a real trained model.

The molecular diagnostic in Table 6 evaluates Eq. (12) from exact individual Hessian-loss gradients. The mean force-score/gradient-norm correlations are 0.57, 0.62 and 0.30 on Test, IRC and NMS, respectively. The corresponding fivefold selection floors are 1.06, 1.15 and 1.67 times Gaussian variance on these diagnostic batches. The proxy is least informative for NMS, where omission leaves the largest exact gradient disagreement. These floors are individually below two; they add to the remaining directional noise rather than alone proving a violation of that threshold. Force-score correlation with gradient norm is useful but insufficient: even perfect norm ordering does not make two gradient vectors parallel. The exact floor measures their full vector disagreement. Likewise, ordering geometry eigenmodes by eigenvalue does not identify corresponding parameter derivatives between different molecular graphs. Arbitrary rotations within degenerate eigenspaces preserve each probe’s isotropy but can alter cross-molecule covariance. The method therefore has no universal alignment guarantee. A separate post-hoc validation check fixes each eigenvector’s largest absolute entry to be positive, using 64 fresh draws on each pilot minibatch. Fivefold coupled variance ratios are 5.01 and 5.32 under this convention, compared with 5.17 and 5.30 for the original phases on the paired draws. Independent fivefold mode probes give 5.14 and 5.23. Thus correcting this sign convention alone does not repair the compression deficit. This check did not alter the frozen training method.

5.2 Failure analysis and transfer beyond the measured regime

For SMC-2, the failed accuracy decisions are Test energy and IRC energy. Their upper bounds are 6.38% and 6.45%, even though the geometric paired changes are only -0.25% and $+0.73\%$,

Table 6: Force-based stratification diagnostics. Reference force medians ($\text{kcal mol}^{-1}\text{\AA}^{-1}$) describe the full source datasets. Each model also contributes an exact 40-molecule Hessian-gradient diagnostic at epoch 80; entries are mean $\pm$ standard deviation over five seeds. The correlation is Spearman’s ρ between force-residual score and exact gradient norm. The SMC-5 selection floor is normalized by independently measured Gaussian gradient variance on these same small batches. These small-batch ratios are mechanism diagnostics, not substitutes for the $B = 400$ audit.

Dataset	Median reference force RMS	Score–gradient ρ	SMC-5 selection floor
Test (RTP)	0.0022	0.57 ± 0.11	1.06 ± 0.03
IRC	3.1695	0.62 ± 0.06	1.15 ± 0.06
NMS	34.8921	0.30 ± 0.10	1.67 ± 0.22

respectively. These are uncertainty-limited failures to establish non-inferiority, not evidence of an energy regression larger than the margin. Test and IRC force and Hessian cells pass, and all three NMS cells pass. The full-allocation Mode-full control also fails the IRC and NMS energy decisions, despite retaining one probe per molecule. Changing the probe law alone therefore does not establish all-cell energy accuracy at this duration and seed count. SMC-5 fails Test energy, Test force and IRC energy, with upper bounds of 5.76%, 5.77% and 9.22%. Its remaining six cells pass, including all three NMS properties. The geometric paired changes in its three failed cells are 0.32%, 0.84% and 2.78%, respectively, again showing why failure of this test should not be presented as demonstrated degradation beyond the margin.

The three evaluation distributions challenge the force proxy differently. RTP configurations are nearly stationary: the median reference force RMS is $0.0022 \text{ kcal mol}^{-1}\text{\AA}^{-1}$, while the median Hessian-entry RMS is $178.36 \text{ kcal mol}^{-1}\text{\AA}^{-2}$. Small physical forces therefore do not indicate small curvature. IRC has a median force RMS of 3.17 with a similar Hessian RMS of 177.52, whereas NMS raises the force median to 34.89 while its Hessian RMS is 167.91. The force scale changes much more than the curvature scale. These distributional measurements explain why a single force-residual ordering is not a calibrated curvature-gradient allocation rule across the three datasets; they are not a causal attribution of any individual accuracy change.

Accuracy and instantaneous Hessian-gradient variance also answer different questions. All configurations retain their energy and force gradients, and the Hessian term carries weight 0.09. A model can consequently have similar finite-horizon prediction errors while failing the conditional Hessian-gradient variance threshold. Conversely, unbiasedness does not make adaptive-optimizer trajectories identical: changing gradient noise changes AdamW’s moment estimates and the distribution of selected checkpoints. A failed non-inferiority bound must be interpreted together with its mean change and uncertainty, rather than as a binary declaration that the model is unusable.

The 80-epoch results cannot establish performance at the published convergence regime. In particular, the reference scheduler’s patience exceeds this duration, so no run benefits from a learning-rate reduction. As training changes both D_i and $\partial_\theta D_i$, it changes the alignment, diagonal noise and selection floor in the variance decomposition. There is no theorem making an early-stage variance ratio persist near convergence. Stage-specific measurements in Table 3, together with the visible published-accuracy gap, limit any transfer claim. Five paired seeds also give limited precision for a stringent all-cell non-inferiority statement. The pooling weights further clarify this limitation: epoch 1 contributes 91.5%, 90.4% and 67.3% of the PHL variance denominator on Test, IRC and NMS. Later stages therefore matter less to the pooled statistic. For SMC-2, the Test ratio rises from 1.60 at epoch 1 to 1.84 at epoch 80, whereas NMS falls from 1.64 to 1.47. Both remain within two at the measured stages, but their different evolution precludes a universal convergence-stage extrapolation.

5.3 Computational cost trade-off

Probe count and wall time separate sharply at this molecular scale. The mean training epoch takes 2.73 s for PHL, 4.22 s for SMC-2 and 3.68 s for SMC-5. The corresponding runtime ratios are 1.55 and 1.35, while the probe reductions are 1.9999 and 4.9998. Figure 6 shows both quantities with the same accounting convention. Full energy and force supervision remains a fixed cost, and a separate

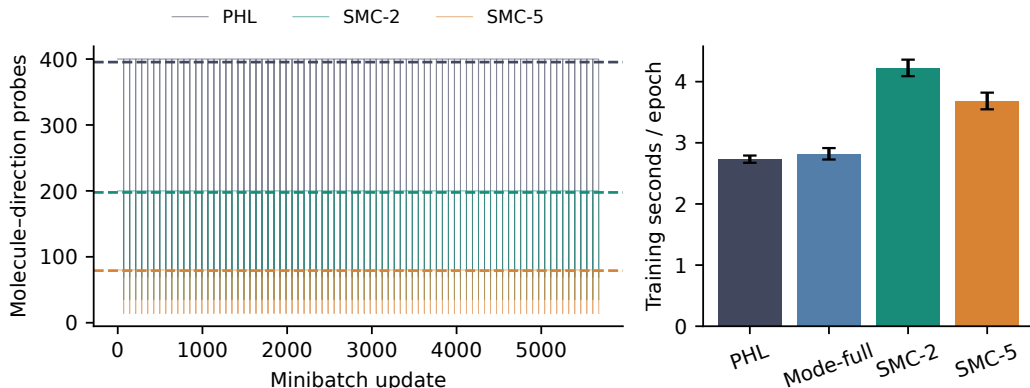


Figure 6: Probe accounting and measured training cost. Left: every minibatch update of the first seed, with dashed time averages; the dips are the final 69-molecule minibatch of each epoch. All five seeds have the same counts. Right: mean epoch time and standard deviation across the five seeds, including data movement, allocation and optimization but excluding validation, audits and disk logging. All timings use the same H200 device.

Table 7: Isolated phase profile (milliseconds per update-sized computation). Phase means and total mean $\pm$ standard deviation over 32 measured repetitions after two warm-ups on the same frozen model and 400-molecule training minibatch. All rows retain the same Hessian-aware data preparation. Each phase has its own synchronization barrier; optimizer updates and disk I/O are excluded. HVP forward includes reconstruction of the selected coordinate graph. The E-F row is a cost-only reference, not a trained accuracy baseline.

Estimator	Data	E/F	Selection	HVP	Backward	Total
E-F only	15.06	5.46	0.00	0.02	6.71	27.25 ± 13.34
PHL	12.62	5.40	0.06	5.73	15.07	38.87 ± 0.14
Mode-full	12.60	5.40	0.06	5.79	15.09	38.94 ± 0.18
SMC-2	12.95	5.47	9.72	11.58	19.79	59.51 ± 1.03
SMC-5	12.82	5.47	4.03	11.21	19.78	53.31 ± 0.83

selected-molecule coordinate graph is required to avoid computing uncounted HVPs on omitted molecules.

The isolated profile in Table 7 separates data transfer, energy/force computation, allocation, HVP construction and backward propagation. It uses a frozen model after all timed training runs have finished, so it avoids overlap with training and diagnostic jobs. The prototype’s force-stratum loop includes device-to-host selection logging and synchronization. Reconstructing the selected graph adds another model forward/force pass. These implementation costs are real parts of this estimator’s measured runtime and cannot be inferred from the HVP count alone. Isolated totals are 38.87 ms for PHL, 59.51 ms for SMC-2 and 53.31 ms for SMC-5. Selection alone costs 9.72 and 4.03 ms for the two reduced variants, while their HVP-forward phases take 11.58 and 11.21 ms versus PHL’s 5.73 ms. Thus the profile independently confirms the absence of a wall-clock speedup. The E-F cost reference retains the same Hessian-aware data preparation and is not an optimized E-F-only pipeline; its larger timing variability is concentrated in that data phase. Vectorizing allocation could reduce this prototype’s classical overhead; it would leave the estimator’s selection-variance floor unchanged.

The input-only RTP eigenbasis cache takes 14.906 s to construct with zero model or reference HVPs. A dense $d \times d$ basis requires $O(d^2)$ storage and a dense eigendecomposition costs $O(d^3)$; rotating each probe also costs $O(d^2)$. These terms make extrapolation to substantially larger systems uncertain. The study’s twenty training runs consume 1.50 GPU-hours of timed updates; the primary gradient audits take 0.44 hours and the full final evaluations take 22.1 minutes. Diagnostics are disclosed separately and do not contribute an update or reduce the reported training budget.

5.4 Scope of the inference

The impossibility statement concerns a universal benefit from nontrivial cross-molecule coupling; it does not rule out successful allocation for particular distributions or better curvature-gradient proxies. The empirical study tests fresh random directions in the randomized-vector regime and uses available full reference Hessians to form target products. It does not establish a result for a dataset containing only one precomputed HVP per molecule. Unbiasedness applies to the conditional loss and its gradient, not to a finite Monte Carlo oracle, a reported square-root RMSE, or the adaptive training trajectory. Equal covariance traces would also not imply equal noise in every parameter direction or equal adaptive-optimizer behavior.

6 Conclusion

Stratified mode coupling is an exactly compensated way to spend fewer curvature probes while retaining full energy and force supervision. Its variance is controlled by two distinct structures: disagreement between molecule gradients and alignment between the differentiated quadratic forms sampled by coupled directions. Neither scalar-loss variance nor probe orthogonality certifies gradient-variance preservation. The exact selection floor and the signed coupling identity explain the conditions under which compression can help and the cases in which it cannot.

In the five-seed, 80-epoch ANI study, nominal twofold and fivefold allocation pass 7 and 6 of nine accuracy tests. Neither establishes the required all-cell non-inferiority, although both pass every NMS accuracy cell. The direct gradient audits and measured computational overhead further distinguish a reduction in HVP count from a preserved optimization process or a runtime improvement. These results concern the tested estimator family and optimization duration, not an impossibility of probe compression for all potentials. A stronger allocation rule must predict curvature-gradient variation well enough to pay for omitted molecules, with its acquisition cost and late-training behavior measured alongside its nominal probe savings.

Reproducibility. The accompanying artifact includes the executed model integration, frozen analysis protocol, all checkpoints and training allocations, full probe archives, per-molecule final errors, and the complete parameter-gradient audits. Tables and figures are generated from these records by `code/aggregate.py` and `code/extra_analysis.py`. `codebase_notes.md` documents reference-code reuse, while `attempts_log.md` records unsuccessful candidates and the precise scope of every retained result.

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Table 8: Validation-only importance-sampling check. $p_i = 0.1/B + 0.9s_i^\alpha/\sum_j s_j^\alpha$, $s_i = \|\tilde{F}_i - F_i\|/(\sqrt{3N_i}N_i^\beta)$. Eighty independent mode-sign probes are drawn with replacement, using compensation $1/(80Bp_i)$, except the full-allocation controls. Entries are conditional gradient-variance ratios from 24 draws. Both columns refer to RTP validation, never final test data.

Estimator	Validation batch 1	Validation batch 2
Gaussian, full allocation	1.00	1.00
Modes, full allocation	0.77	0.77
$\alpha = 0.0, \beta = 0.0$	5.24	6.04
$\alpha = 0.5, \beta = 0.0$	4.83	5.51
$\alpha = 0.5, \beta = 1.0$	4.48	5.17
$\alpha = 1.0, \beta = 0.0$	4.95	5.12
$\alpha = 1.0, \beta = 1.0$	4.53	4.72
$\alpha = 2.0, \beta = 0.0$	6.09	5.48
$\alpha = 2.0, \beta = 1.0$	6.24	6.48

A Scope of the exploratory candidates

The validation-only pilot includes independent Gaussian, Cartesian sign, and mode-basis sign probes; full-allocation and reduced-allocation Hadamard coupling; uniform and force-stratified selection; and factors two, three, and five. Figure 4 reports all these candidates. The threefold and fivefold candidates did not satisfy the pilot’s gradient variance threshold. The fivefold candidate was nevertheless retained for the full five-seed study to examine whether downstream accuracy and instantaneous gradient variance tell the same story. The threefold candidate has no claimed final training result. These distinctions and the implementation reconciliation are recorded in `attempts_log.md`; no unsuccessful candidate is silently relabeled as a completed accuracy comparison.

A separate validation-only importance-sampling diagnostic uses the same pilot checkpoint and a 10% uniform mixture to keep every molecule’s probability positive. Each of P draws contributes $\ell_i(v)/(PBp_i)$, whose expectation is the full minibatch Hessian loss. The probabilities are detached functions of the available force residual; they do not use Hessian-gradient measurements. This is an adaptation of standard importance sampling, motivated by Katharopoulos and Fleuret [2018], rather than a reproduction of that work’s neural-network gradient-norm bound. Table 8 retains every tested exponent. It did not alter the frozen primary estimator or final-seed evaluation protocol.